HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Analyzed equipment and process performance data using statistical methods to identify trends and outliers in semiconductor manufacturing, driving corrective actions that reduced variation by **15%**.
- Troubleshot complex issues in metrology toolsets, developing scalable solutions and implementing strategies that prevented recurrence of critical failures.
- Optimized tool performance and recipes through sustaining engineering activities, enhancing defect detection capabilities by **10%**.
- Supported recipe development, optimization, and transfers to ensure high-sensitivity defect detection, working with cross-functional teams to refine detection parameters.
- Led equipment improvement initiatives with cross-functional teams to enhance the operational performance of defect metrology systems, influencing stakeholders to drive change.
- Developed and maintained Python scripts for data extraction, validation, and analysis of equipment performance metrics, supporting data-driven decision-making for yield optimization.
- Containerized data analysis tools and services using **Docker** and deployed them onto **AWS EC2** instances, improving deployment consistency and reducing setup time for new analysis environments.

SKILLS

Semiconductor Manufacturing – Defect Metrology, Yield Analysis, Process Improvement, Equipment Engineering, Recipe Development

Data Analysis & Visualization – JMP, Statistical Analysis, Data Extraction, Data Validation, Trend Analysis, Outlier Detection

Programming & Scripting – Python, SQL, Shell Scripting

Cloud & Containerization – AWS (EC2, S3, CloudWatch), Docker

Engineering Disciplines – Electrical Engineering, Mechanical Engineering, Materials Science, Mathematics

PROJECTS

Semiconductor Yield Optimization Platform

- Developed Python scripts to automate the extraction and aggregation of defect data from multiple metrology tool sources, processing approximately **50,000** defect records daily.
- Implemented statistical analysis modules using pandas and NumPy to identify root causes of yield loss, contributing to a **12%** reduction in critical defect types.
- Built interactive dashboards using **Power BI** to visualize yield trends and equipment performance metrics for manufacturing stakeholders, facilitating faster issue identification.

Automated Equipment Performance Monitoring

- Engineered a monitoring solution using Python and AWS CloudWatch to track key performance indicators for critical metrology equipment, alerting on deviations from optimal parameters.
- Designed and implemented SQL queries against PostgreSQL to extract historical equipment logs and fault data, enabling detailed root cause analysis for recurring issues.
- Collaborated with operations teams to define alert thresholds and reporting requirements, ensuring the monitoring system directly addressed their production needs.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)